

Exp No: 14

EMBEDDED C PROGRAM TO DISPLAY "HELLO WORLD" MESSAGE

Program:

```
#include <reg51.h>; #include <stdio.h>;

void main (void){

    SCON = 0x50; /* SCON: mode 1, 8-bit UART, enable rcvr */ TMOD = 0x20; /* TMOD:
    timer 1, mode 2, 8-bit reload */ TH1 = 0xFD; /* TH1: reload value for 9600 baudrate */

    TR1 = 1; /* TR1: timer 1 run */

    TI = 1; /* TI: set TI to send first char of UART */ while (1)

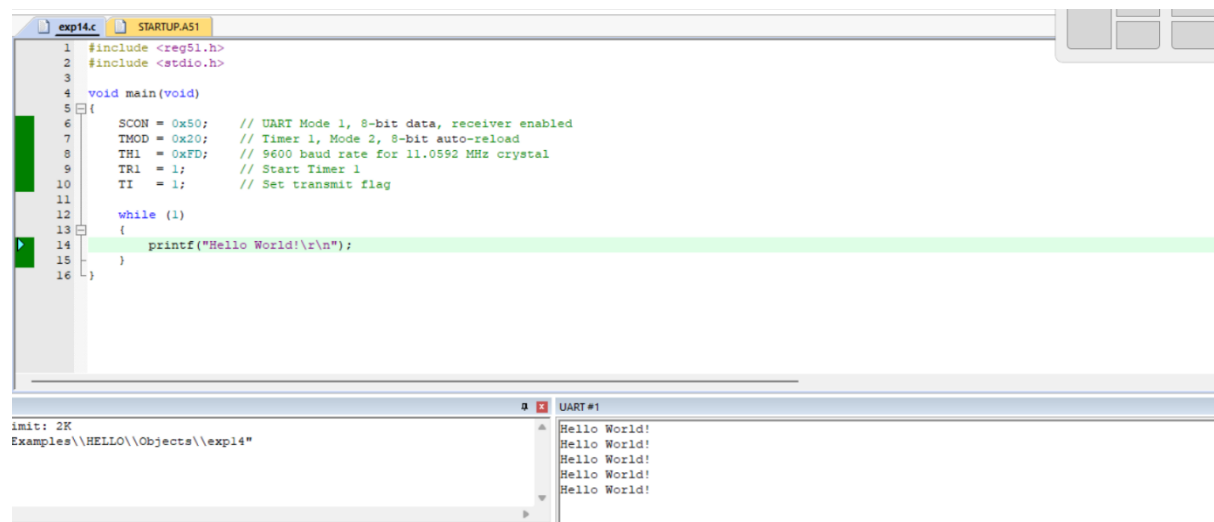
    {

    printf (&"Hello World ! \n &");

    }

}
```

OUTPUT:



The screenshot shows an IDE with two main windows. The top window, titled 'exp14.c', contains the C program code. The code is as follows:

```
1 #include <reg51.h>
2 #include <stdio.h>
3
4 void main(void)
5 {
6     SCON = 0x50; // UART Mode 1, 8-bit data, receiver enabled
7     TMOD = 0x20; // Timer 1, Mode 2, 8-bit auto-reload
8     TH1 = 0xFD; // 9600 baud rate for 11.0592 MHz crystal
9     TR1 = 1; // Start Timer 1
10    TI = 1; // Set transmit flag
11
12    while (1)
13    {
14        printf("Hello World!\n");
15    }
16 }
```

The bottom window, titled 'UART#1', shows the output of the program, which is 'Hello World!' repeated five times on separate lines. The status bar at the bottom left indicates 'init: 2K' and 'Examples\HELLO\Objects\exp14'.